Marking Rubric (Summary)

- Data preprocessing and handling: 4
- Model implementation and comparison: 7
- Evaluation protocol and metrics: 4
- Clarity and reproducibility: 3
- Report polish and visualization quality: 2

4 Variational Autoencoder (VAE) [35 Marks]

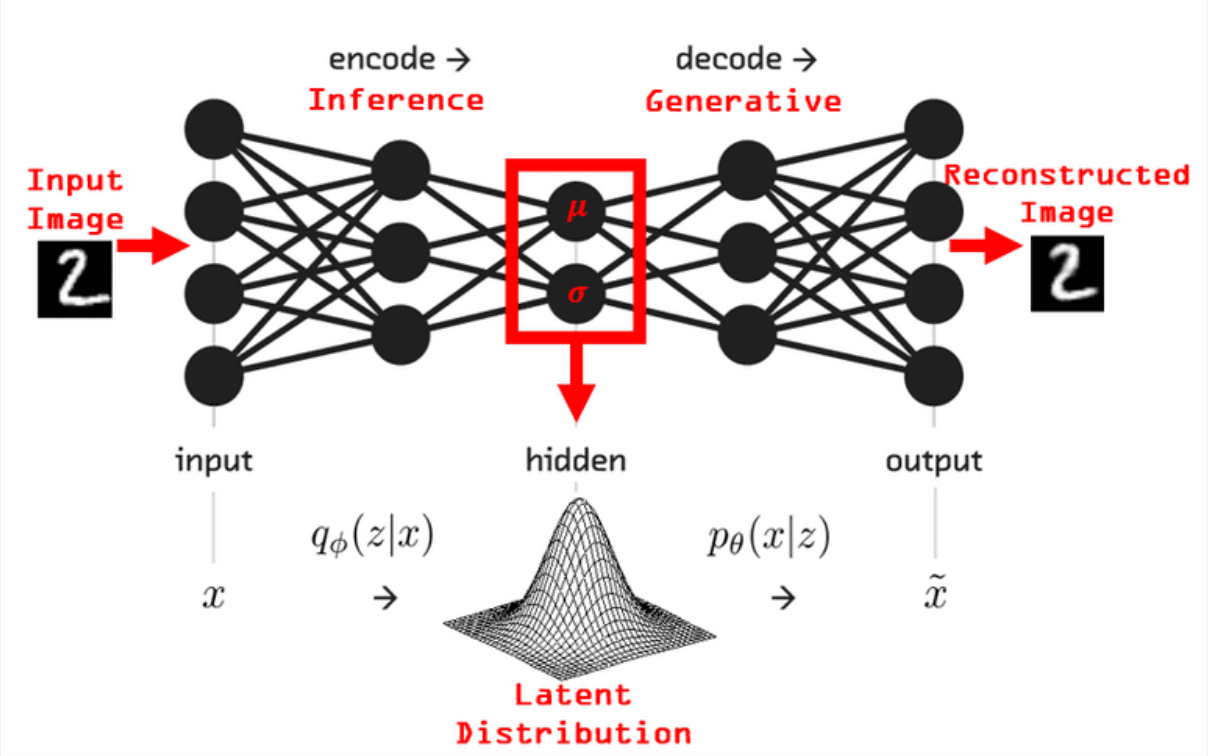


Figure 2: Variational Autoencoder Architecture

4.1 Introduction

A Variational Autoencoder (VAE) is a generative deep learning model that learns to encode data into a continuous latent space and decode it back to reconstruct the input. In this task, you will implement and analyze a VAE trained on the **Fashion-MNIST** dataset. You are allowed to use external libraries for this question.

4.2 Dataset Preparation [2 Marks]

Load and preprocess the **Fashion-MNIST** dataset using PyTorch utilities.

- Normalize the data and create DataLoader instances for both training and testing.
- Choose suitable batch size and normalization strategy.

4.3 Model Architecture [10 Marks]

Design and implement a Variational Autoencoder consisting of:

1. **Encoder:** A neural network mapping the input image to a latent representation defined by a mean vector μ and a log-variance vector $\log(\sigma^2)$.
2. **Reparameterization Trick:** Use $z = \mu + \epsilon \cdot \sigma$ with $\epsilon \sim \mathcal{N}(0, I)$ to enable stochastic sampling with gradient backpropagation.
3. **Decoder:** A neural network reconstructing images from latent samples.

Experiment with different latent dimensions and network depths, and discuss how these changes influence the smoothness of the latent space and reconstruction fidelity.

4.4 Loss Function [4 Marks]

The total loss for the VAE is a combination of:

- **Reconstruction Loss:** Measures pixel-wise similarity between input and reconstruction using Binary Cross-Entropy (BCE).
- **KL Divergence:** Encourages the latent distribution to approximate a standard normal prior.

Formally, the objective function is given by:

$$\mathcal{L} = \mathbb{E}_{q(z|x)}[\log p(x|z)] - D_{KL}(q(z|x) \,||\, p(z))$$

Explain the role of each term in shaping the learned latent representation.

4.5 Training Procedure [2 Marks]

Train your model using an optimizer of your choice (e.g., Adam).

- Plot the total loss and its components over epochs.

4.6 Experimental Analysis [9 Marks]

In this section, you will conduct systematic experiments to understand the influence of key parameters on VAE performance.

- Vary the weighting between reconstruction loss and KL divergence using a β-parameter (as in β-VAE). Create a gif/video showing how the latent space evolves for values of $\beta = [0.1, 0.5, 1]$ for 3 classes of your choosing, make sure to color-code the data points in the latent space according to their class to observe clustering in the latent space (An example video has been attached in the dataset folder). Discuss how increasing β affects image sharpness/quality, Diversity, Cluster separation etc.
- Summarize your findings in a short table comparing quantitative metrics (e.g., reconstruction loss) and qualitative observations.

4.7 Evaluation and Visualization [3 Marks]

- Display a set of original and reconstructed test images side-by-side.
- Generate new samples by sampling from $\mathcal{N}(0, I)$ in the latent space and decoding them.
- Interpret the quality, diversity, and realism of generated samples, measure the Fréchet Inception Distance (FID) to evaluate goodness of newly generated samples.

4.8 Effect of Frozen Latent Parameters [5 Marks]

Investigate the influence of fixed latent distribution parameters on generation quality.

- Freeze the mean μ as 0 and try $\sigma = [0.1, 0.5, 1]$
- Generate new samples by decoding these frozen latent vectors.
- Compare the generated samples to those obtained from the standard stochastic sampling ($z = \mu + \epsilon \cdot \sigma$).
- Evaluate and discuss the effect of freezing μ and/or σ on image diversity, sharpness, and representational smoothness in the latent space.